Smart Crop Disease Detection System — Project Report

Name: Hanusha K

1. Abstract

Agriculture forms the backbone of economies worldwide, yet crop diseases silently destroy between 20 and 40 percent of global yield every single year. The Smart Crop Disease Detection System addresses this critical challenge by combining deep learning with edge computing to build an intelligent, offline-capable system that automatically identifies plant diseases from leaf images in real time. Using a hierarchical MobileNetV2-based neural network trained on over 54,000 images from the PlantVillage dataset, the system achieves a validation accuracy of 93.58 percent across 38 disease classes spanning 14 crop species. The trained model is converted to TensorFlow Lite format and deployed on a Raspberry Pi 4, where it operates alongside a DHT22 temperature and a Pi Camera Module V2 to create a fully autonomous, field-deployable disease detection unit.

2. Introduction

2.1 The Problem

Manual crop inspection is one of the most labor-intensive processes in modern agriculture. A single hectare of farmland can contain between 10,000 and 15,000 plants, translating to anywhere from 500,000 to 1.5 million individual leaves that must be examined. With only two to three workers, this process can take three to five days per field inspection cycle. Beyond the sheer time cost, there exists a significant expertise gap — even experienced farmers struggle to correctly identify diseases when more than 38 distinct conditions can affect their crops, many of which present with visually similar symptoms. A wrong diagnosis leads directly to wrong treatment, which wastes money, damages the environment through unnecessary chemical application, and ultimately results in preventable yield loss

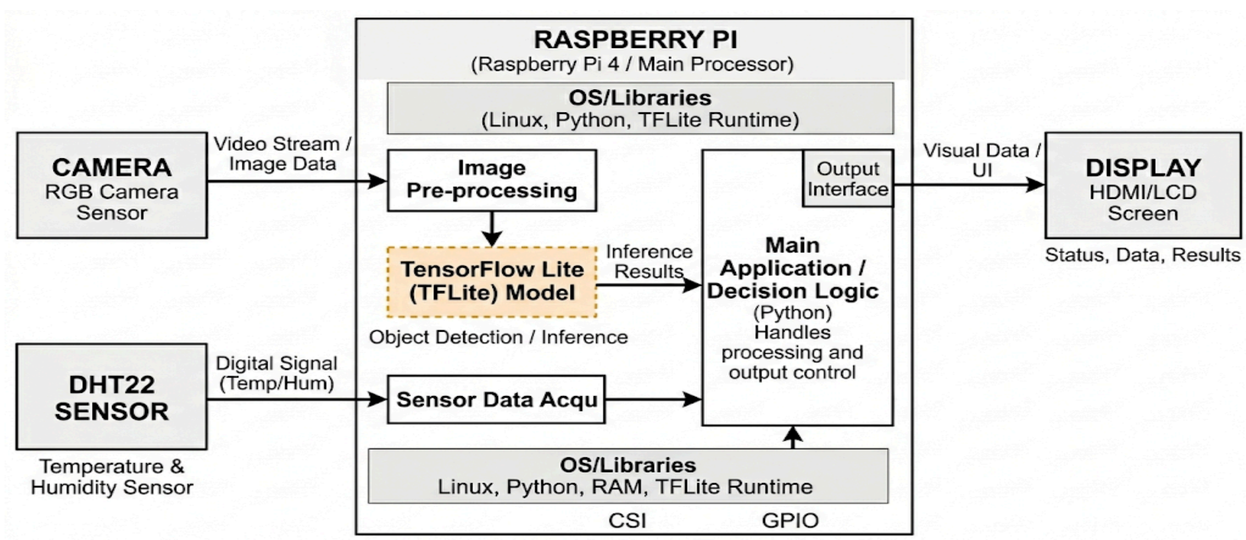
Chemical treatments applied without precise diagnosis increase costs by a factor that could otherwise be reduced by 60 to 70 percent with targeted intervention.

Proposed Solution

This project proposes an AI-powered edge computing system that captures leaf images through a camera module, processes them locally on a Raspberry Pi, and delivers a precise disease diagnosis within seconds — completely offline. The system simultaneously reads environmental conditions such as temperature and humidity from a DHT22 sensor, correlating atmospheric data with disease risk levels to provide the farmer with a richer, more actionable report.

System Architecture Overview

The system is built on three interconnected layers working together seamlessly. The first is the sensing layer, which consists of the Raspberry Pi Camera Module V2 capturing high-resolution leaf images and the DHT22 sensor continuously monitoring temperature and humidity. The second is the processing layer, where a hierarchical deep learning model runs locally on the Raspberry Pi using TensorFlow Lite to classify both the plant species and the disease present. The third is the output layer, where results are displayed on a 3.5-inch TFT LCD touchscreen with LED indicators — green for a healthy plant and red when a disease is detected.



Hardware Components

4.1 Main Processing Unit

The Raspberry Pi 4 with 2GB or 4GB of RAM serves as the central processing unit of the system. It was chosen for its excellent balance between computational performance and cost, its built-in WiFi and Bluetooth capabilities, and its exceptionally strong community support. Critically, MobileNetV2 was designed with devices like the Raspberry Pi in mind, making this hardware pairing a natural fit.

Camera Module

The Raspberry Pi Camera Module V2 provides 8-megapixel resolution with 1080p video capability. Its native integration with Raspberry Pi OS through the Picamera2 library eliminates compatibility complications. The camera connects via a 15-centimeter ribbon cable directly to the CSI camera port on the Pi. It captures the leaf image at close range, providing the model with a clean, well-lit input suitable for accurate classification

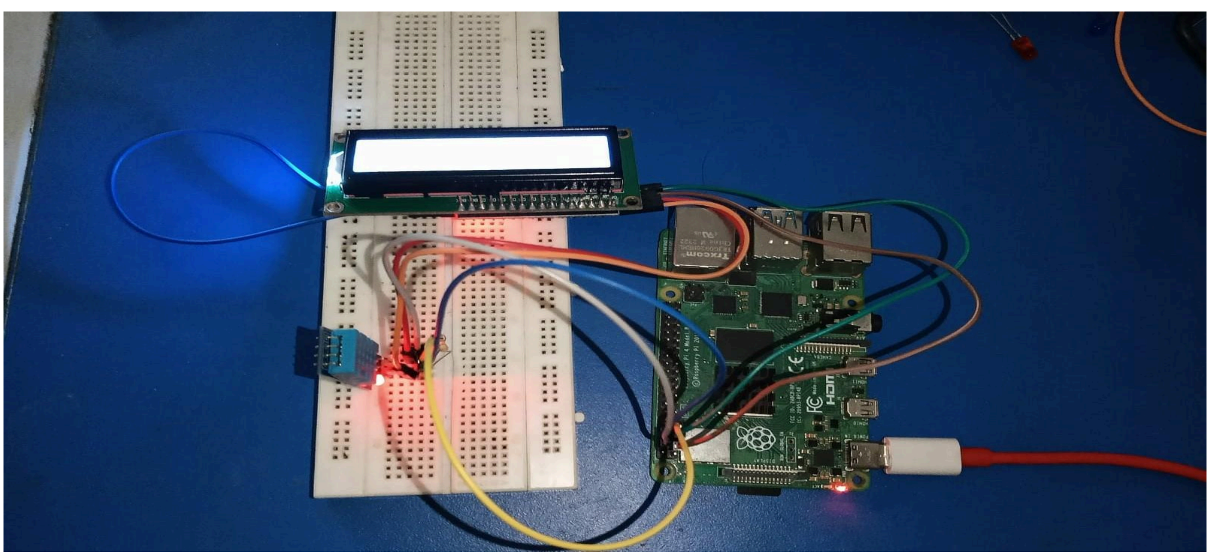
Temperature Sensor : The DHT22 sensor measures temperature from -40°C to 80°C with ±0.5°C accuracy.These readings are incorporated into the output report to alert the farmer when environmental conditions are conducive to disease spread — for example, high temperature.

Display

A 3.5-inch TFT LCD touchscreen with 480×320 pixel resolution displays the prediction results, confidence scores, plant identification, and sensor readings in a user-friendly interface

Supporting Components

The system also uses a 32GB or 64GB MicroSD card for the operating system and model files, a dedicated Raspberry Pi 4 power supply, a breadboard with connecting wires, resistors for the LED circuit, and a power adapter suitable for field use.



Software Stack

The entire software pipeline runs on Raspberry Pi OS (64-bit), the official operating system for Raspberry Pi hardware. Python 3.8 or higher serves as the programming language throughout. TensorFlow Lite handles inference at the edge, providing the lightweight runtime environment needed to run the converted model efficiently without the full TensorFlow framework. OpenCV is used for any image preprocessing steps before the image is fed to the model. NumPy and Pandas manage array operations and data handling. Picamera2 provides the camera control interface for capturing real-time images from the Pi Camera Module.

Dataset

The PlantVillage dataset serves as the primary training data source, containing 54,306 images across 38 disease categories covering 14 distinct crop species. The crops included span the most economically important plants in global agriculture: Tomato, Potato, Corn (Maize), Apple, Grape, Pepper (Bell), Strawberry, Peach, Cherry, Blueberry, Orange, Raspberry, Soybean, and Squash. Images are provided in JPG format at 256×256 pixel resolution and are labeled with both healthy and diseased conditions.

Machine Learning Model

The system employs a custom Hierarchical Two-Branch Architecture built upon a MobileNetV2 backbone. MobileNetV2 was selected for its efficiency on edge devices, utilizing depthwise separable convolutions and inverted residual blocks to maintain a lightweight 14 MB footprint suitable for real-time inference on Raspberry Pi hardware. To leverage transfer learning, the backbone was initialized with ImageNet weights and frozen, ensuring robust feature extraction while minimizing training overhead.

To resolve visual ambiguities between similar disease patterns across different species, the architecture splits into two functional branches. The Plant Branch classifies the specimen into one of 14 plant classes, while the Disease Branch utilizes a concatenation layer to combine raw image features with the output of the Plant Branch. This "conditioning" ensures that disease identification is contextually informed by the plant species, preventing cross-species misclassification. Both branches utilize **He** Normal initialization, Batch Normalization, and Dropout to ensure stable gradient flow and effective regularization across all dense layers.

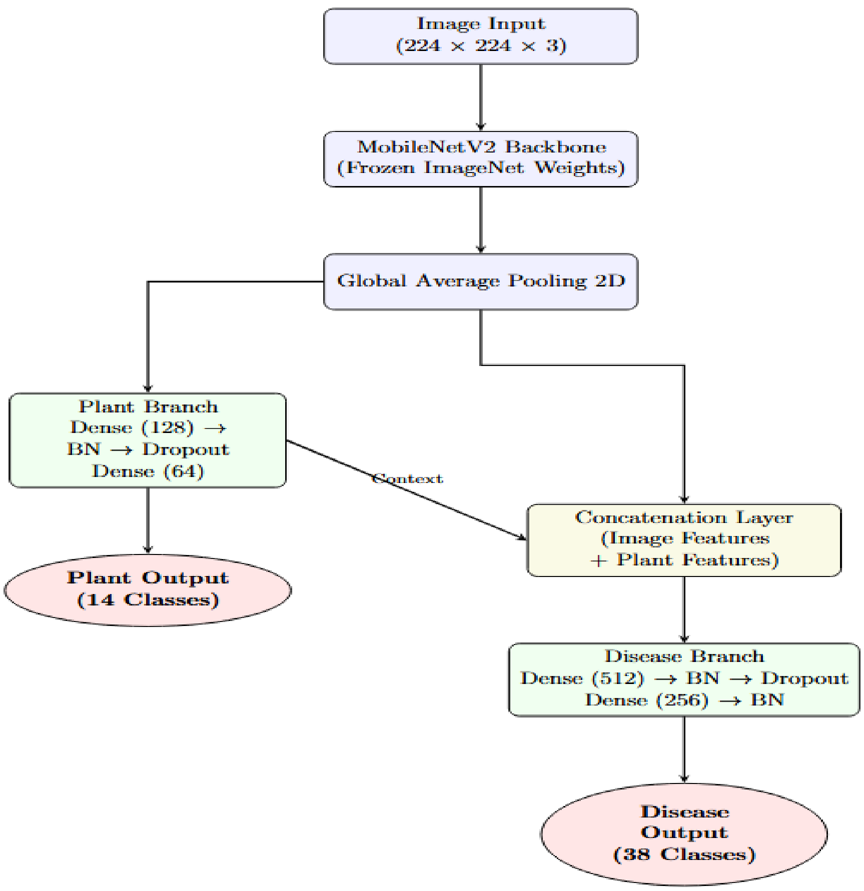


Figure 1: Hierarchical Two-Branch Model Architecture

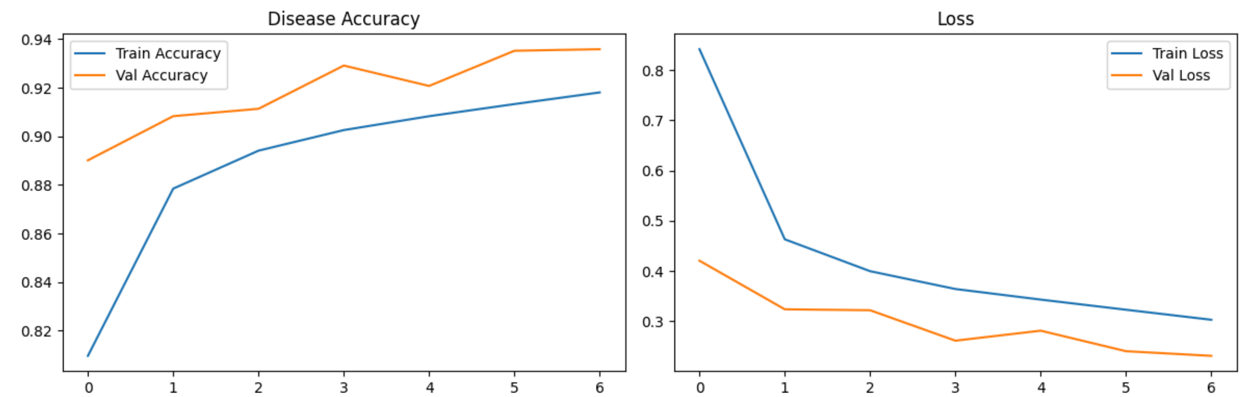
Training Results

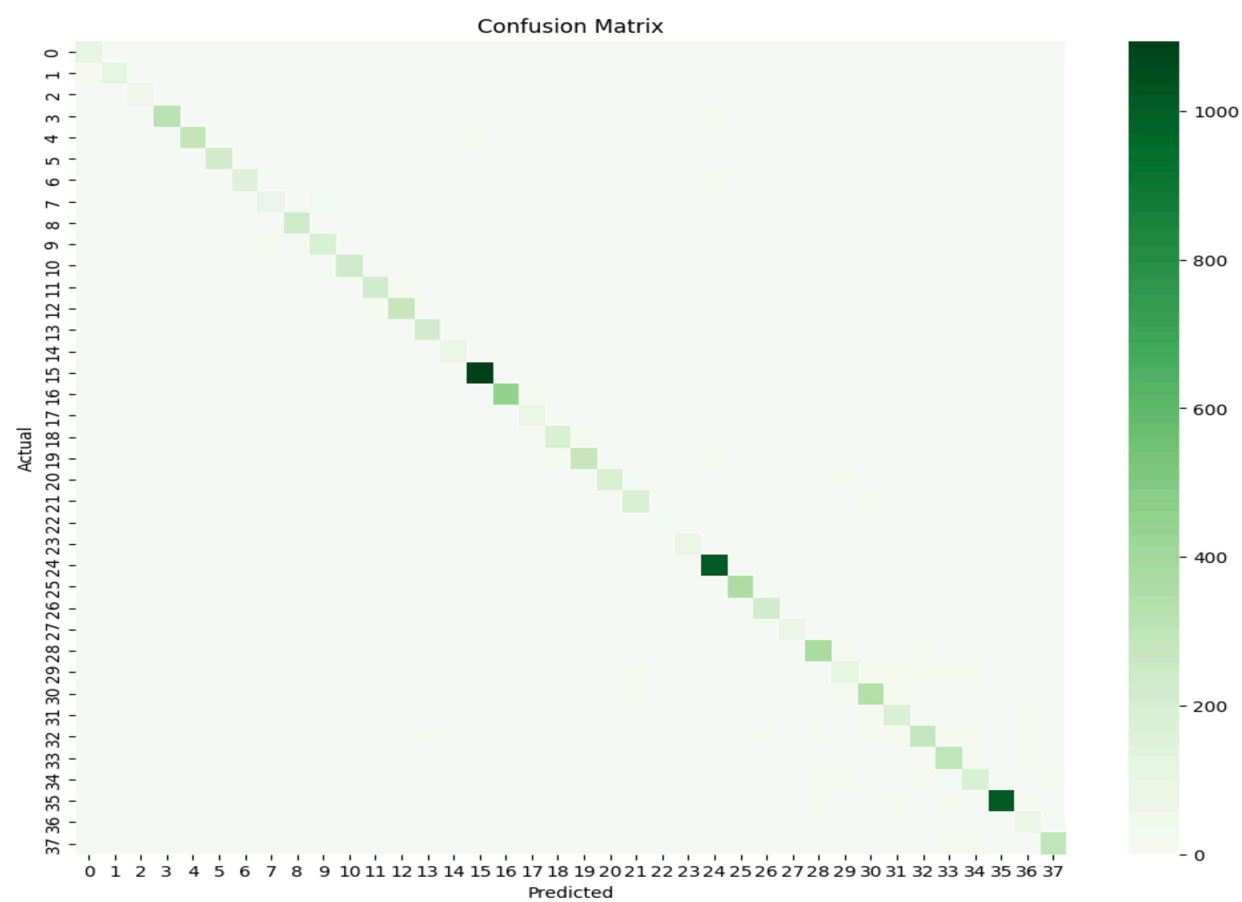
Epoch-by-Epoch Progression

The model demonstrated consistent and steady improvement across all seven epochs. Epoch one established a strong baseline with a validation disease accuracy of 89.01 percent. By epoch two this improved to 90.83 percent, and by epoch three to 91.13 percent. Epoch four showed a more significant jump to 92.91 percent, while epoch five showed a slight regression to 92.07 percent — the only epoch where the model did not save a new best checkpoint. Epoch six recovered strongly to 93.52 percent, and the final epoch seven produced the best result of 93.58 percent, which was saved as the definitive model.

Final training disease accuracy reached 91.81 percent while validation disease accuracy reached 93.58 percent — with validation slightly exceeding training accuracy, which is a healthy sign attributable to Dropout being active during training but disabled during validation, and to the augmented training images being harder to classify than the clean validation images.

Plant identification accuracy reached 97.56 percent by the final epoch, confirming that the Plant Branch learned to reliably identify which of the 14 crop species it was examining before the Disease Branch made its disease-level prediction.





Classification Performance

9.1 Overall Metrics

The model achieved an overall accuracy of 94 percent on the 10,849-image validation set. The macro average F1-score was 0.92 and the weighted average F1-score was 0.94, indicating consistently strong performance across both large and small classes.

9.2 Best Performing Classes

Several disease classes achieved near-perfect classification. Apple Cedar Apple Rust achieved an F1-score of 0.99 with perfect recall. Orange Haunglongbing (Citrus Greening) achieved 0.99, benefiting from the largest support in the dataset at 1,101 images. Corn Common Rust achieved 0.98, Grape Leaf Blight achieved 0.99, Grape Healthy achieved 0.99, Squash Powdery Mildew achieved 0.99, Strawberry Leaf Scorch achieved 0.98, Peach Bacterial Spot achieved 0.98, and Tomato Yellow Leaf Curl Virus achieved 0.97 with 1,071 support images. These classes benefit from visually distinctive patterns and, in many cases, larger training sets.

Challenging Classes

Tomato Early Blight was the most difficult class, achieving an F1-score of only 0.68. This is largely because early-stage blight symptoms on tomato leaves are visually very similar to Late Blight and Target Spot, which are themselves frequently confused. Corn Cercospora Leaf Spot achieved 0.75, as its grey rectangular lesions resemble Northern Leaf Blight lesions closely. Tomato Leaf Mold achieved 0.80, Tomato Septoria Leaf Spot achieved 0.82, and Potato Healthy achieved 0.82 — the latter suffering from a very small support of only 30 validation images, limiting the model's exposure to this class.

Following training, the model was converted from the `.keras` format to **TensorFlow Lite (.tflite)**. By applying **default quantization**, the model size was reduced from **17.12 MB to approximately 4–5 MB** (a 70% reduction). This optimization is critical for the Raspberry Pi’s ARM processor, ensuring rapid inference without the need for a dedicated GPU.

The edge deployment package consists of four essential components:

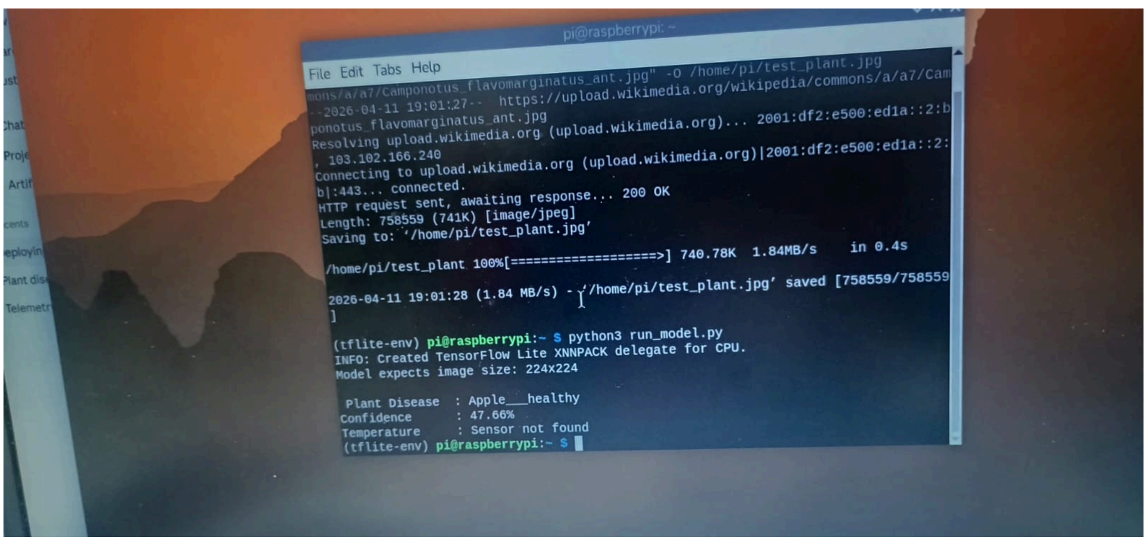
- **`plant_disease.tflite`**: The optimized model weights.
- **`class_names.json`**: Mapping for the 38 disease labels.
- **`plant_names.json`**: Mapping for the 14 plant species.
- **`predict.py`**: The Python script managing the pre-processing and dual-branch inference pipeline.

Raspberry Pi Implementation

Software Deployment

The deployment pipeline on the Raspberry Pi installs three key Python packages: tflite-runtime for running the TFLite model, Pillow for image loading and resizing, and numpy for array operations. The TFLite interpreter loads the model and allocates tensors once at startup. For each leaf image submitted — whether from the stored image folder or captured live by the camera — the inference pipeline preprocesses the image to 224×224 pixels, normalizes pixel values to the 0–1 range, and runs the model to obtain both disease and plant output tensors.

The prediction logic then applies the hierarchical filtering at inference time as well. The plant branch output identifies which plant species is present. The disease branch output is then masked to zero out all disease classes that do not belong to the identified plant, and the remaining probabilities are renormalized. This ensures the final answer always presents a plant and disease from the same species — a biologically meaningful guarantee that no single-output classifier can provide.



Conclusion

The Smart Crop Disease Detection System successfully demonstrates that advanced artificial intelligence can be made accessible, affordable, and deployable in the agricultural environments where it is needed most. By combining MobileNetV2 transfer learning with a novel hierarchical two-branch architecture, the system achieves 93.58 percent validation accuracy across 38 disease classes while ensuring that plant and disease predictions are always biologically consistent. The TensorFlow Lite conversion brings this capability to the Raspberry Pi 4, enabling offline edge inference that requires no internet connectivity — a critical requirement for deployment in remote farming areas.

The economic value of such a system is substantial. Early and accurate disease detection can save between 15 and 25 percent of total crop yield that would otherwise be lost. Replacing three to five days of manual inspection labor with automated continuous monitoring saves significant costs every single week. Targeted treatment based on accurate diagnosis reduces chemical usage by 60 to 70 percent, lowering both costs and environmental impact. The system's ability to detect disease when only one or two leaves are affected — before the condition spreads through the field — is arguably its most valuable capability.

This project demonstrates that low-cost hardware like the Raspberry Pi, combined with carefully designed deep learning models and thoughtful system architecture, is capable of delivering professional-grade agricultural intelligence to farmers anywhere in the world.